THE DAD'S GUIDE

How an Aerospace Engineer Can Help His Daughter Become a Competitive Clinical Research Data Analyst

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A companion guide to the 8-Week Skill-Building Roadmap

Your background: C/C++ embedded systems, STM32, UAVCAN, MAVLink, 60 GitHub repos (github.com/geosmall), Claude Code power user

Includes: Claude Desktop & Cowork Safety Guide + VM Sandbox Setup

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YOUR SUPERPOWER (AND YOUR BLIND SPOTS)

Let's be honest about what you bring to the table. You're an experienced engineer who writes C for bare-metal STM32 targets, contributes to open-source flight controller projects, and uses Claude Code 10+ hours a week. You think in registers, protocols, and build systems. That's a very different world from clinical trial databases and Tableau dashboards — but the underlying engineering mindset transfers beautifully.

✓ Where You Can Lead

Git & GitHub — you have 60 repos and real PR experience. You can teach this in your sleep.

Python — you know C/C++, so Python's syntax will feel like cheating. You can learn pandas alongside her.

Claude / AI tools — you're already a Claude Code power user. You understand prompt engineering intuitively.

Debugging mindset — reading error messages, systematic troubleshooting, and "read the docs" culture.

Environment setup — installing tools, managing paths, configuring editors. This trips up beginners constantly.

⚠ Where You'll Learn Together

SQL — you probably haven't written a JOIN since college (if ever). That's OK. It's learnable in days.

Tableau — drag-and-drop visual analytics is a different paradigm from code-first engineering.

Clinical research domain — she knows this, you don't. Let her be the expert here.

REDCap — clinical data capture tool. Completely new to both of you.

HIPAA & compliance — she lives this; you'll need to understand the guardrails to give good advice.

The best dynamic here: you're not her teacher — you're her engineering mentor and technical pair-programmer. She brings the domain knowledge (clinical research, HIPAA, what the data means). You bring the engineering habits (version control, clean code, systematic debugging, environment setup). Where neither of you knows the content (SQL, Tableau), you learn side by side.

YOUR ROLE, WEEK BY WEEK

Here's what you can specifically do each week to help without taking over. The key rule: help her build the skill, don't build it for her.

WEEKS 1–2: SQL Foundations

Your SQL level: Probably rusty or new. That's fine — SQL is more like writing English questions than writing C.

What you can do:

- **Set up her environment.** Install SQLite or PostgreSQL locally. This is where most beginners waste hours and get frustrated. You can do it in 15 minutes and save her the pain.
- **Download and import the Synthea dataset together.** Use Claude Code to write the import scripts. This is a great early win that shows her how Claude accelerates setup tasks.
- **Learn alongside her.** Do the same SQLBolt exercises. Compare your approaches. You'll pick up JOINs fast with your engineering logic — but let her work through them at her pace.
- **Initialize her GitHub repo.** Walk her through creating clinical-research-portfolio, writing the first README, setting up .gitignore, and making her first commit. This is YOUR domain — make it feel easy, not ceremonial.

WEEKS 3–4: Tableau + Statistics

Your Tableau level: Probably zero. That's actually fine — Tableau is designed for non-programmers. She may outpace you here.

What you can do:

- **Install Tableau Public for her** and help her connect it to the SQL database or CSV files from Weeks 1–2.
- **Be her audience, not her teacher.** When she builds a dashboard, look at it and tell her what's clear and what's confusing. You're the non-clinical person in the room — if you can understand the dashboard, it's working.
- **Help with the statistics portion.** You have an engineering math background. Help her connect concepts like standard deviation and confidence intervals to practical meaning. Use Claude to generate practice problems.
- **Don't over-engineer it.** Your instinct will be to script everything and automate the pipeline. Resist. Tableau is point-and-click on purpose. Let her learn the tool the way it's designed.

WEEK 5: Python + REDCap

Your Python level: You write C for STM32. Python will feel like pseudocode to you. This is YOUR week to shine.

What you can do:

- **Set up her Python environment properly.** Anaconda or venv, pip, VS Code with Python extension. Make sure it works before she touches a tutorial. Environment hell kills motivation.

- **Pair-program the first pandas script.** Sit together, load a CSV, and explore it. Show her df.head(), df.describe(), df.isnull().sum(). Let her drive the keyboard while you explain what's happening.
- **Use Claude Code to build scaffolding.** Show her how you'd use Claude Code to generate a data cleaning script, then walk through what it produced and modify it together. This is a real-world skill demonstration.
- **Step back for REDCap.** REDCap is a clinical domain tool. Let her explore it independently using the training videos. Ask her to show you what she built and explain why the form logic matters for data quality.

WEEK 6: AI Tools & Agents

Your AI level: You're a Claude Code power user doing 10+ hours a week. You're probably ahead of most professionals here.

What you can do:

- **Teach prompting by example.** Show her your real Claude Code sessions. Demonstrate how you iterate on prompts, provide context, and refine outputs. She'll learn more from watching you work with Claude for 20 minutes than from any tutorial.
- **Help her build a multi-step workflow.** Your engineering instinct for breaking problems into steps is exactly what agent-style AI workflows need. Help her design a chain: summarize dataset → suggest questions → write SQL → interpret results.
- **Model healthy skepticism.** Show her that you verify Claude's outputs. In clinical research, an unchecked AI hallucination could have real consequences. Teach the "trust but verify" habit.
- **Reinforce HIPAA boundaries hard this week.** This is where the temptation to paste real work data into AI tools is strongest. Make sure she understands: synthetic data only, always.

WEEK 7: GitHub & Portfolio

Your GitHub level: Pull Shark x2, Arctic Code Vault Contributor, 60 repos. This is your home turf.

What you can do:

- **Code-review her repo structure.** Look at her folder organization, README quality, and .gitignore completeness the way you'd review a PR on one of your open-source projects. Give constructive feedback.
- **Teach branching through a real exercise.** Have her create a feature branch, make changes, and submit a PR to her own repo. Then you review and merge it. This simulates real collaborative workflows.
- **Security audit her repos.** With your open-source experience, you know what should never be in a public repo. Do a thorough check for any data files, hardcoded paths, API keys, or anything that could be sensitive. This is a genuine safety contribution.
- **Help her write a great profile README.** You know what makes a GitHub profile look professional. Help her create one that highlights her clinical research focus and links to her best work.

WEEK 8: Capstone Project

Your role: Project advisor and QA reviewer. She drives. You help her think through architecture, catch bugs, and polish the final deliverable.

What you can do:

- **Help her plan the pipeline end-to-end.** Whiteboard the data flow: source → SQL queries → Python cleaning → Tableau viz → AI-assisted report. Your systems engineering thinking is exactly what this needs.
- **Review her final code and documentation.** Read her SQL, Python scripts, and READMEs. Does it make sense? Is it commented? Could a hiring manager follow it? Give her the same level of feedback you'd give an open-source contributor.

- **Do a final security sweep.** Before she shares her portfolio with anyone, go through every file in every repo. No data leaks, no credentials, no personally identifiable information. This matters.
- **Celebrate the win.** She just built a full data analytics portfolio from scratch. That's a big deal.

CLAUDE CODE: YOUR SECRET WEAPON

You're already using Claude Code extensively for your embedded work. Here are specific ways to leverage it to help her learning — and to model effective AI-assisted workflows she can adopt for her own career.

1. Scaffold her projects

Use Claude Code to generate project templates: folder structures, .gitignore files, README templates with placeholders she fills in, and boilerplate Python scripts. Show her the prompts you used so she learns the pattern.

2. Generate practice datasets

Ask Claude Code to create synthetic clinical trial CSV files with realistic columns (patient_id, site_id, enrollment_date, treatment_arm, ae_severity, etc.) and intentional data quality issues for her to find. This gives her unlimited practice material tailored to her exact learning stage.

3. Build her a local SQL playground

Use Claude Code to set up a SQLite database with multiple related tables mimicking a clinical trial data model. Write the schema, import scripts, and a few starter queries. Give her the database and let her explore. You build the infrastructure; she builds the queries.

4. Create "challenge" exercises

Use Claude Code to generate SQL and Python challenges at the right difficulty level: "Write a query that finds all patients who had an adverse event within 30 days of enrollment at sites with more than 50 participants." Include the answer key in a separate file.

5. Demonstrate AI-assisted debugging

When her SQL query returns unexpected results or her Python script throws an error, show her how you'd paste the error into Claude Code and work through the fix. Narrate your thought process: "I'd ask Claude to explain the error, but I'd also read the traceback myself first." This teaches the workflow, not just the answer.

THE ENGINEER-DAD TRAPS (AND HOW TO AVOID THEM)

You know your own tendencies. Here are the ones to watch out for:

The Trap	Why It's Tempting	What to Do Instead
Solving it for her	You see the SQL bug in 10 seconds. She's been stuck for 20 minutes.	Ask guiding questions: "What does the error say? What table are you joining on?" Let her find it.
Over-engineering the setup	You want Docker, Makefiles, CI/CD, and a proper dev environment.	Keep it simple. SQLite + VS Code + GitHub. She needs to learn SQL, not DevOps.
Dismissing Tableau	"Why not just write a Python script?" Because her coworkers and managers use Tableau.	Respect that different tools serve different audiences. Her stakeholders don't read Python.
Going too fast	You process technical content quickly. She's learning a new domain AND new tools.	Match her pace. If a "week" takes two weeks, that's success, not failure.
Skipping the clinical context	You want to get to the technical part. The clinical stuff seems like background.	The domain knowledge IS the career advantage. Ask her to explain why enrollment tracking matters.
Forgetting she's the expert too	You're used to being the senior engineer in the room.	She knows HIPAA, clinical workflows, and what data means. Defer to her on those topics.

YOUR PRE-WORK: GET READY BEFORE WEEK 1

Spend 2–3 hours on this before you start with her. It'll make everything smoother.

1. **Install SQLite and a GUI tool** (DB Browser for SQLite is free and beginner-friendly). Verify it works on her machine.
2. **Download the Synthea dataset** (synthea.mitre.org). Use Claude Code to write an import script that loads the CSVs into SQLite tables.
3. **Install Python (Anaconda distribution)** and verify pandas, matplotlib, and jupyter work. Install VS Code with the Python and Jupyter extensions.
4. **Install Tableau Public** (free from tableau.com). Open one of the sample datasets to make sure it launches.
5. **Do the first 10 SQLBolt exercises yourself** (sqlbolt.com). It'll take you 30 minutes and give you enough SQL to be a helpful partner in Weeks 1–2.
6. **Read one overview of HIPAA basics** (the HIPAA Journal has good summaries). You need to understand what PHI is and why it can never go into public tools. Ask her to quiz you — she'll enjoy that.
7. **Create her GitHub repo structure** using Claude Code. Set up the folders, .gitignore, and a starter README. Make her first commit a collaboration: she writes the intro paragraph, you handle the Git commands.
8. **Set up her Claude Desktop safety sandbox** (see the "Protecting Her Machines" section below). At minimum, create a Claude_Workspace folder on both machines. Optionally, set up a VM together using UTM (Mac) or Hyper-V (Win11) for full isolation.

PROTECTING HER MACHINES: CLAUDE DESKTOP & COWORK

Claude Desktop and Cowork are agentic tools — they can read, write, and delete files on her machine with her user-level permissions. Her instinct not to blindly trust them is good engineering thinking. Here's a layered defense strategy, from simplest to most robust.

Layer 1: Dedicated Workspace Folder (Do This First)

The single most effective protection and it takes five minutes. Create a folder like ~/Claude_Workspace (Mac) or C:\Claude_Workspace (Windows). Only grant Claude access to this folder. Copy files in when she wants Claude to work on them, copy results out when done. Her Documents, Desktop, Downloads, and everything else stays completely untouched. Anthropic's own safety documentation recommends exactly this approach.

Layer 2: Current Backups

Before she starts using Cowork for the first time, verify Time Machine (Mac) or File History (Win11) is on and current. On Mac, plug in an external drive or set up a network destination. On Windows, enable File History in Settings or set up OneDrive folder sync. This is her safety net for any "oops" scenario, not just AI-related.

Layer 3: Understand the Permission Prompts

Cowork requires explicit permission before permanently deleting any files. She'll see a prompt and must select "Allow" before Claude can delete anything. The key habit: always read what Claude is asking to do before clicking Allow. Watch for scope creep — is Claude accessing files or websites she didn't mention? If something feels off, stop the task immediately.

Layer 4: Start Simple, Add Capabilities Later

- **Skip the Chrome extension initially.** Web content is the primary vector for prompt injection attacks. She doesn't need it for learning projects.
- **Don't install third-party MCPs or plugins yet.** Each extension expands Claude's capabilities and introduces new attack surface. Stick to verified extensions from the Claude Desktop directory only.
- **Skip scheduled tasks until she's comfortable.** Scheduled tasks run when she's not watching. Start with interactive-only use where she can see what Claude is doing.
- **Never process untrusted files.** Don't ask Claude to summarize a random PDF from an email. Hidden instructions in documents can trick AI agents into unexpected actions.
- **Keep Claude Desktop updated.** Multiple high-severity security vulnerabilities have been patched in 2025–2026. Running an outdated version means running with known vulnerabilities.

Layer 5: The VM Sandbox (Your Weekend Project Together)

For maximum peace of mind, run Claude Desktop inside a lightweight virtual machine. This gives complete isolation — Claude literally cannot access the host machine's files. This is the belt-and-suspenders option: overkill for most learning projects, but it eliminates the "what if" worry entirely and makes a great father-daughter engineering project.

💻 macOS: UTM (Free, Open Source)

UTM is a free, open-source VM app for Mac that runs beautifully on Apple Silicon. It can run macOS, Windows, and Linux guests.

Setup Steps:

9. **Download UTM from mac.getutm.app** (free) or from the Mac App Store ($9.99 for auto-updates, same software).
10. **Create a macOS VM** (easiest option on Apple Silicon): UTM > New > Virtualize > macOS. Download the IPSW restore image when prompted. Allocate 4+ GB RAM and 60+ GB disk.
11. **Or create an Ubuntu Linux VM** (lighter weight): UTM > New > Virtualize > Linux. Download Ubuntu Desktop ARM64 ISO. This uses fewer resources and Claude Desktop runs on Linux.
12. **Install Claude Desktop inside the VM.** Download from claude.com/download within the VM's browser. Sign in with her Claude account.
13. **Use UTM's shared folder feature** to pass files between the host and VM. Create a single shared folder (e.g., ~/VM_Shared) that acts as the bridge. Files she wants Claude to work on go in there; results come back out the same way.
14. **Take a snapshot after setup.** UTM supports VM snapshots. Once everything is installed and working, save a snapshot. If anything goes wrong, she can revert to a clean state in seconds.

Pro tip: A macOS VM on Apple Silicon M-series chips runs near-native speed. She won't feel like she's using a VM for normal tasks. A Linux VM uses even less resources and is plenty for Claude Desktop + Python + SQL work.

🖥️ Windows 11: Hyper-V (Built-In, Free)

Hyper-V is built into Windows 11 Pro and Enterprise (not Home). If she has Win11 Home, she can use VirtualBox (free) as an alternative.

Setup Steps:

15. **Enable Hyper-V:** Settings > Apps > Optional Features > More Windows Features > check "Hyper-V" > restart. Or run in PowerShell (admin):
    Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V -All
16. **Create a Windows 11 VM:** Open Hyper-V Manager > Quick Create > select Windows 11 dev environment (free from Microsoft, pre-configured). Or download a Windows ISO and create a new VM manually. Allocate 4+ GB RAM.
17. **Or create an Ubuntu VM** (lighter): Quick Create > Ubuntu. One-click setup, uses fewer resources, Claude Desktop has a Linux version.
18. **Install Claude Desktop inside the VM.** Download from claude.com/download. For a Windows guest, download the Windows installer. For Ubuntu, download the Linux version.

19. **Set up Enhanced Session Mode** for shared clipboard and folder access between host and VM. In Hyper-V Settings, enable Enhanced Session Mode. Create a shared folder for file transfer.
20. **Create a checkpoint (snapshot).** Right-click the VM > Checkpoint. This saves the VM's exact state. She can revert anytime if something goes sideways.

Win11 Home alternative: If her Windows machine runs Home edition, download Oracle VirtualBox (free, virtualbox.org). The setup is similar: create a VM, install an OS, install Claude Desktop inside it, use shared folders for file transfer.

Which Approach Should She Start With?

Comfort Level	Recommendation	Protection Level
"Just keep it safe"	Dedicated folder + backups + read permission prompts carefully	Covers 95% of realistic risk. Good enough for learning projects.
"I want a safety net"	All of the above + skip Chrome ext + no third-party MCPs + interactive only	Very safe. She sees everything Claude does before it happens.
"I want zero risk to my machine"	Run Claude Desktop in a VM (UTM or Hyper-V). Everything stays inside the sandbox.	Maximum isolation. Claude literally cannot touch host files. Snapshot = instant undo.

Our recommendation: Start with the dedicated folder approach (10 minutes to set up). If she feels cautious, set up a VM together as a weekend project — it's a useful skill to have anyway, and you'll have it done in under an hour. Once she sees how the permission model works and gains confidence, she can decide whether to keep using the VM or run Claude natively.

The best part of this plan?

You get to build something together.

Not a drone firmware. Not a flight controller. A career.